

Instructions of `multireflects`

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1 About

`multireflects` is a `Python` library designed for generating multi-band reflection-dominated light curves, with computational acceleration implemented in `C`. It can be integrated with MCMC sampling libraries, such as `emcee`, to explore the orbital parameters of binary systems dominated by reflection effects.

The main advantages of `multireflects` are:

- (1) A single execution can simultaneously produce light curves in multiple passbands.
- (2) For the same input parameters, its runtime performance (excluding initialization time) is at least one order of magnitude faster than commonly used reflection light-curve generators such as `Phoebe`, and on average about two orders of magnitude faster.
- (3) The stellar surface is divided into a fine mesh (over 10,000 surface elements), resulting in smooth light curves.

The main limitations are:

- (1) The use of ellipsoidal approximations and relatively coarse surface gravity interpolation can introduce significant errors when modeling ellipsoidal-variation-dominated light curves, particularly in regions where gravity darkening is strong.
- (2) Users must properly configure a `C` compiler and possess some basic knowledge of compilation to successfully install and use the library.

2 Definitions

- (1) In the parameter naming convention, quantities with the suffix “1” refer to the primary star, while those with the suffix “2” correspond to the companion star.
- (2) The zero phase of the light curve is defined within the orbital plane as the moment when, after the companion passes through the ascending node, the angle formed by the companion–orbital focus–ascending node reaches $\frac{\pi}{2}$.

3 Filter Object

```
1 passband_obj = multireflects.passband(T, wave, I_lambda1, S_lambda,  
2   I_lambda2)  
3 passband_obj.A1 = ...  
4 passband_obj.A2 = ...  
5 passband_obj.u1 = ...  
6 passband_obj.u2 = ...
```

A filter (passband) object can be constructed by calling `multireflects.passband`, which returns an instance named `passband_obj`. The input arguments are defined as follows. `T` is a one-dimensional `numpy.float32` array representing the effective temperature grid in units of Kelvin. `wave` is a one-dimensional `numpy.float32` array representing the wavelength grid in units of angstroms. `S_lambda` is a one-dimensional `numpy.float32` array with length `len(wave)`, whose values lie in the range [0,1]. It represents the transmission curve of the filter. `I_lambda1` is a two-dimensional `numpy.float32` array with shape `(len(wave), len(T))`, describing the spectral flux distribution of the primary star as a function of wavelength for different effective temperatures. The argument `I_lambda2` has the same structure and meaning as `I_lambda1`, but corresponds to the companion star.

To ensure correct execution, several additional attributes must be manually assigned to `passband_obj`. The attribute `passband_obj.A1` represents the reflection coefficient (Bond albedo) of the primary star in this passband; in most cases, this can be taken to be equal to the global reflection coefficient. Similarly, `passband_obj.A2` corresponds to the reflection coefficient of the companion star. The attribute `passband_obj.u1` is a one-dimensional `numpy.float32` array of length 4, representing the four-parameter limb-darkening coefficients of the primary star in this passband. The specific functional form of the limb-darkening law and the definition of these coefficients can be found in the works of Antonio Claret. Likewise, `passband_obj.u2` has the same structure and meaning as `passband_obj.u1`, but applies to the companion star.

4 Light Curve Workflow

4.1 Initialization

```
1 lc_obj = multireflects.lc(phi, passbandnum)
```

A light-curve workflow object can be created by calling `multireflects.lc`, which returns an instance named `lc_obj`. In the initialization arguments, `phi` is a one-dimensional `numpy.float32` array with values in the range [0,1], representing the orbital phase of the light curve. The parameter `passbandnum` specifies the number of passbands (filters) included in the calculation.

4.2 Input Orbital Parameters

You need to manually enter the following parameters to ensure that the program runs correctly.

- `lc_obj.T1` The effective temperature of the primary star in unit of Kelvin.
- `lc_obj.T2` The effective temperature of the companion star in unit of Kelvin.
- `lc_obj.r1` The stellar radii of the primary star in unit of $R_{\odot}$.
- `lc_obj.r2` The stellar radii of the companion star in unit of $R_{\odot}$.
- `lc_obj.A1` The bolometric reflection coefficient (filters independent) of the primary star, with value constrained to the range [0,1].
- `lc_obj.A2` The bolometric reflection coefficient (filters independent) of the companion star, with value constrained to the range [0,1].
- `lc_obj.gdc1` The gravity darkening coefficient of the primary star, within [0,1]. In general, a value of 1 is appropriate for stars with radiative envelopes, while a value of 0.32 is typically adopted for stars with convective envelopes.
- `lc_obj.gdc2` The gravity darkening coefficient of the companion star, within [0,1].
- `lc_obj.u1` The linear limb-darkening coefficient of the primary star, with value constrained to the range [0,1]. The corresponding limb-darkening law and coefficient definitions can be found in the works of Antonio Claret.

`lc_obj.u2` The linear limb-darkening coefficient of the companion star, with value constrained to the range $[0,1]$.

`lc_obj.incl` The orbital inclination in degrees, with values ranging from 0° to 90° .

`lc_obj.q` The mass ratio which is defined as the companion mass divided by the primary mass.

`lc_obj.P` The orbital period in unit of days.

`lc_obj.sma` The sum of the semi-major axes of the two stars' orbits in unit of $R_\odot$.

`lc_obj.ecc` The orbital eccentricity, within $[0,1]$.

`lc_obj.w2` The longitude of periastron for the companion star in radians, with values in the range $[0, 2\pi)$.

`lc_obj.RV_gamma` The systemic radial velocity of the binary system in unit of km/s.

4.3 Generate the Light Curve

```

1 passbandlist = [passband_obj, ...]
2 lc_obj.lc_mode0(passbandlist)
3 lc_obj.lc_mode1(passbandlist)

```

Multiple filter (`passband_obj`) instances can be combined into a `Python` list named `passbandlist`. By calling `lc_obj.lc_mode0(passbandlist)` or `lc_obj.lc_mode1(passbandlist)`, the light curves can be generated. The resulting light curves are stored in `lc_obj.lc`, which is a `numpy.float32` array of shape `(passbandnum, len(phi))`, representing the light curves in each passband.

The method `lc_obj.lc_mode0` accounts for mutual reflection between the stars, including both the primary's reflection on the companion and the companion's reflection on the primary. In contrast, `lc_obj.lc_mode1` neglects the reflection of the primary on the companion to accelerate the computation. It should be noted that neither `lc_obj.lc_mode0` nor `lc_obj.lc_mode1` considers higher-order (multiple) reflections.